Rectangular channel Couette-Poiseuille flow

$$\rho D_t \vec{v} = \rho[\underbrace{\partial_t \vec{v}}_{\text{steady state} \to \emptyset} + (\vec{v}\cdot\nabla)\vec{v}] = -\nabla p + \eta \nabla^2 v + \underbrace{\beta\eta\nabla(\nabla\cdot\vec{v})}_{\text{incompressible fluid} \to \emptyset} + \overbrace{\rho\vec{g} + \rho_{el}\vec{E}}^{\text{Ignore}} \to \emptyset$$

flow in direct of $\hat{e}_x \to \vec{v} = (0,0,v_x) \to v_x \partial_x(\vec{v})$

translational invariance along x direction $\to v = v(y,z)$

$\to \partial_x(\vec{v}) = \emptyset$

$\Rightarrow -\nabla p + \eta \nabla^2 v = \emptyset \Rightarrow \eta(\partial_y^2 + \partial_z^2)v_x = \partial_x p(x)$ $\left(\begin{array}{l} v_y = \emptyset \to \partial_y p = \emptyset \\ v_z = \emptyset \to \partial_z p = \emptyset \end{array}\right)$ $\Rightarrow p = p(x)$

$\frac{d}{dx}p(x) = c_1 \to p(x) = c_1 x + c_2$ $\quad p(0) = p^* + \Delta p$, $p(L) = p^*$

$p(0) = c_2 = p^* + \Delta p$, $p(L) = c_1 L + p^* + \Delta p = p^* \to c_1 L + \Delta p = \emptyset \to c_1 = -\frac{\Delta p}{L}$

$\Rightarrow p(x) = -\frac{\Delta p}{L}x + \Delta p + p^* \to \boxed{p(x) = \frac{\Delta p}{L}(L-x) + p^*}$ $\quad *: \partial_x p(x) = -\frac{\Delta p}{L}$

$\to \eta(\partial_y^2 + \partial_z^2)v_x = -\frac{\Delta p}{L} \to (\partial_y^2 + \partial_z^2)v_x = -\frac{\Delta p}{\eta L}$ Poisson equation.

linear partial differential 2nd order equation

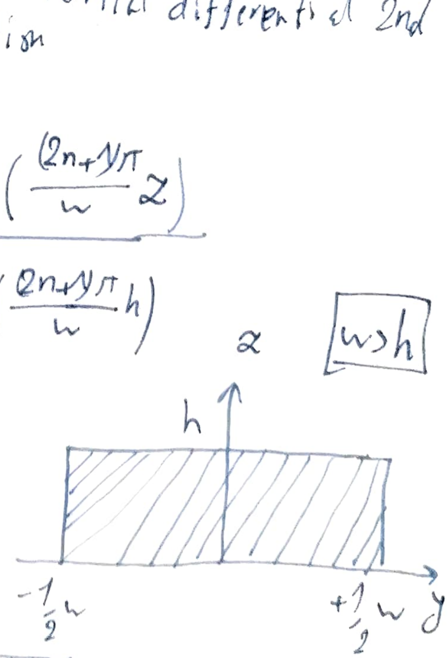
linearity: $\underline{v_x = v_x^{\text{Couette}} + v_x^{\text{Poiseuille}}}$

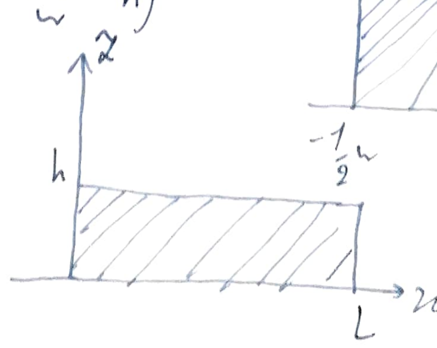
(I) Couette flow

$$v_x^c(y,z) = \sum_{n=0}^{\infty} \frac{4 v_{wall}}{(2n+1)\pi} \cdot (-1)^n \cdot \cos\left(\frac{(2n+1)\pi}{w} y\right) \cdot \frac{\sinh\left(\frac{(2n+1)\pi}{w} z\right)}{\sinh\left(\frac{(2n+1)\pi}{w} h\right)}$$

$$Q = \sum_{n=0}^{\infty} \frac{8 v_w w^2}{(2n+1)^3 \pi^3} \cdot (-1)^{n+1} \cdot \frac{\cosh\left(\frac{(2n+1)\pi}{w} h\right) - 1}{\sinh\left(\frac{(2n+1)\pi}{w} h\right)}$$

$\boxed{w > h}$





① Poiseuille flow

$$v_x(y,z) = \frac{\Delta P}{2\eta L}(hz - z^2) - \sum_{n=0}^{\infty} \frac{4\Delta P h^2}{\eta L \pi^3 (2n+1)^3} \cdot \frac{\cosh\left(\frac{(2n+1)\pi z}{h}\right)}{\cosh\left(\frac{(2n+1)\pi w}{2h}\right)} \cdot \sin\frac{(2n+1)\pi z}{h}$$

$$Q = \frac{\Delta P}{12\eta L} \cdot h^3 w \left[1 - \sum_{n=0} \frac{192\, h}{w \pi^5 (2n+1)^5} \cdot \tanh\left(\frac{(2n+1)\pi w}{2h}\right)\right]$$

$$\Rightarrow v_x = v_{x,\text{couette}} + v_{x,\text{Poiseuille}}, \quad Q = Q_{\text{couette}} + Q_{\text{Poiseuille}}$$

$$v_x = \sum_{n=0}^{\infty} \left[\frac{4 v_w}{(2n+1)\pi} \cdot (-1)^n \cdot \cos\left(\frac{(2n+1)\pi}{w} y\right) \cdot \frac{\sinh\left(\frac{(2n+1)\pi}{w} z\right)}{\sinh\left(\frac{(2n+1)\pi h}{w}\right)}\right\} \text{couette}$$

$$- \frac{4\Delta P h^2}{\eta L \pi^3 (2n+1)^3} \cdot \frac{\cosh\left(\frac{(2n+1)\pi y}{h}\right)}{\cosh\left(\frac{(2n+1)\pi w}{2h}\right)} \cdot \sin\left(\frac{(2n+1)\pi z}{h}\right)\Bigg] + \frac{\Delta P}{2\eta L}(hz - z^2)$$

Poiseille

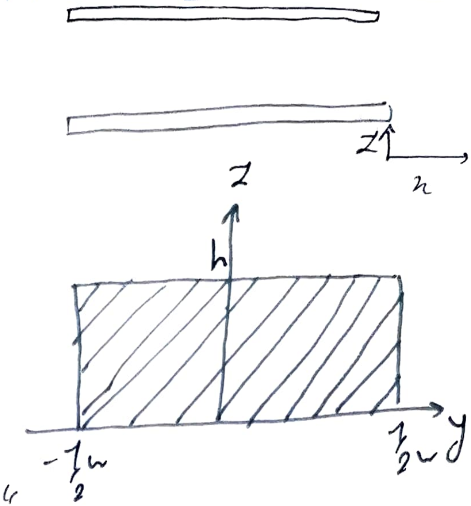
$$Q = \sum_{n=0} \frac{8 v_w w^2}{(2n+1)^3 \pi^3} \cdot (-1)^{n+1} \cdot \frac{\cosh\left(\frac{2n+1}{w}\pi h\right) - 1}{\sinh\left(\frac{2n+1}{w}\pi h\right)}$$

$$+ \frac{\Delta P}{12\eta L} h^3 w \left(1 - \sum_{n=0} \frac{192 h}{w \pi^5 (2n+1)^5} \tanh\frac{(2n+1)\pi w}{2h}\right)$$

* Couette flow (I)

$(\partial_y^2+\partial_z^2)\, v_x = -\frac{\Delta P}{\eta L} \to \Delta P = \emptyset$

$\Rightarrow (\partial_y^2+\partial_z^2)\, v_x = \emptyset$ Laplace equation



no-slip boundary condition
$$\begin{cases} v_x(z=0, y) = \emptyset \\ v_x(y, z=h) = v_{wall} \\ v_x(y=\pm\frac{1}{2}w, z) = \emptyset \end{cases}$$

$v_x = Y(y)Z(z) \Rightarrow (\partial_y^2+\partial_z^2)v_x = Y''(y)Z + YZ'' = \emptyset$

$Y''Z = -YZ'' \Rightarrow \frac{Y''}{Y} = -\frac{Z''}{Z} = -\lambda^2 \Rightarrow Y'' + \lambda^2 Y = \emptyset$

$(D^2+\lambda^2)Y = \emptyset \Rightarrow D = \pm i\lambda$

$\Rightarrow Y = A\cos\lambda y + B\sin\lambda y$

$\frac{Z''}{Z} = \lambda^2 \Rightarrow Z'' - \lambda^2 Z = \emptyset \Rightarrow (D^2 - \lambda^2)Z = \emptyset \Rightarrow D = \pm\lambda$

$\Rightarrow Z = C\cosh\lambda z + D\sinh\lambda z$

$\Rightarrow v_x(y,z) = (A\cos\lambda y + B\sin\lambda y)(C\cosh\lambda z + D\sinh\lambda z)$

symmetry for $y \to v(y,z) = v(-y,z) \to B = \emptyset$

$v_x = (A\cos\lambda y)(C\cosh\lambda z + D\sinh\lambda z)$

$v_x(y=\pm\frac{w}{2}, z) = \emptyset \Rightarrow A\cos(\pm\lambda\frac{w}{2}) = \emptyset$ $\quad A \neq \emptyset \to \cos(\pm\lambda\frac{w}{2}) = \emptyset$

$\frac{\lambda w}{2} = \frac{(2n+1)\pi}{2} \Rightarrow \lambda_n = \frac{(2n+1)\pi}{w}$

$v_x(y, z=0) = \emptyset \Rightarrow C + 0 = \emptyset \Rightarrow C = \emptyset$

$v_x = \sum_n (A_n\cos\lambda_n y)(D_n\sinh\lambda_n z) \xrightarrow{A_nD_n = C_n'} v_x(y,z) = C_n'\cos\lambda_n y\,\sinh\lambda_n z$

$v_x(y, z=h) = \sum_n C_n'\cos\lambda_n y\,\sinh\lambda_n h = v_{wall} \to v_{wall} = \sum_n \underbrace{C_n'\sinh(\lambda_n h)}_{\alpha_n}\cos\lambda_n y$ → Fourier Cosine series

$\alpha_n = \frac{2}{w}\int_{-w/2}^{w/2} v_{wall}\cos\lambda_n y\,dy = \frac{2}{w}v_{wall}\cdot\frac{1}{\lambda_n}\sin\lambda_n y\Big|_{-w/2}^{w/2} = \frac{2}{w}v_{wall}\cdot\frac{1}{\lambda_n}\left(\sin\lambda_n\frac{w}{2} - \sin -\frac{\lambda_n w}{2}\right) = \frac{2}{w}\cdot v_w\cdot\frac{1}{\lambda_n}\cdot 2\sin\frac{\lambda_n w}{2}$

$\Rightarrow \alpha_n = 4\frac{v}{w\lambda_n}\sin\frac{\lambda_n w}{2}$

$\sin\frac{\lambda_n w}{2} = \sin\frac{(2n+1)\pi\cancel{w}}{2\cancel{w}} = (-1)^n \to \alpha_n = \frac{4v}{\lambda_n w}(-1)^n$

$$\alpha_n = c'_n \sinh(\lambda_n h) = \frac{4 v_w}{\lambda_n w} (-1)^n \to c'_n = \frac{4 v_w}{\lambda_n w} \cdot \frac{(-1)^n}{\sinh(\lambda_n h)}$$

$$\to v_x(y,z) = \sum_{n=0}^{\infty} \frac{4 v_w}{\lambda_n w} \cdot \frac{(-1)^n}{\sinh(\lambda_n h)} \cdot \cos(\lambda_n y) \sinh(\lambda_n z)$$

$$\lambda_n = \frac{(2n+1)\pi}{w}$$

$$= \sum_{n=0}^{\infty} \frac{4 v_w}{(2n+1)\pi} \cdot (-1)^n \cdot \cos\left(\frac{(2n+1)\pi}{w} y\right) \frac{\sinh\left(\frac{(2n+1)\pi}{w} z\right)}{\sinh\left(\frac{(2n+1)\pi}{w} h\right)}$$

$$Q = \int_0^h dz \int_{-w/2}^{w/2} dy \, v_x(y,z)$$

$$\int_{-w/2}^{w/2} dy \cos \frac{(2n+1)\pi}{w} y = \frac{w}{(2n+1)\pi} \sin\left(\frac{(2n+1)\pi}{w} y\right)\Bigg|_{-w/2}^{w/2}$$

$$= \frac{w}{(2n+1)\pi} \left((-1)^n + (-1)^n\right)$$

$$= \frac{2w}{(2n+1)\pi} (-1)^n$$

$$\int_0^h dz \sinh\left(\frac{(2n+1)\pi}{w} z\right) = \frac{w}{(2n+1)\pi} \cosh\left(\frac{(2n+1)\pi}{w} z\right)\Bigg|_0^h$$

$$= \frac{w}{(2n+1)\pi} \cdot \left(\cosh \frac{(2n+1)\pi}{w} h - 1\right)$$

$$\Rightarrow Q = \sum_{n=0}^{\infty} \frac{4 v_w w}{\pi(2n+1)} \cdot (-1)^n \cdot \frac{2w}{(2n+1)\pi} \cdot (-1)^n \cdot \frac{\cosh \frac{(2n+1)\pi h}{w} - 1}{\sinh \frac{(2n+1)\pi}{w} h} \cdot \frac{w}{(2n+1)\pi}$$

$$= \sum_{n=0}^{\infty} \frac{8 v_w w^2}{(2n+1)^3 \pi^3} \cdot (-1)^{n+1} \cdot \frac{\cosh\left(\frac{2n+1}{w} \pi h\right) - 1}{\sinh\left(\frac{2n+1}{w} \pi h\right)}$$

* Poiseuille flow (II)

$(\partial_y^2+\partial_z^2)\, v_x(y,z) = -\frac{\Delta P}{\eta L}$ $\quad v_x(y=\pm\frac{w}{2}, z)=\emptyset \quad v_x(y, z=0)=\emptyset$

$v_x(y, z=h)=\emptyset$

$v_x = v_0 + v_c$

$v_0 \equiv$ answer for infinite parallel plate case

$v_0 = \frac{-\Delta P}{2\eta L} z(h-z) \to$ answer

$(\partial_y^2+\partial_z^2)(v_0+v_c) = -\frac{\Delta P}{\cancel{2}\eta L} \to (\partial_y^2+\partial_z^2)v_c + \partial_z^2\left(-\frac{\Delta P}{2\eta L}(zh-z^2)\right) = -\frac{\Delta P}{\eta L}$

$(\partial_y^2+\partial_z^2)v_c - \frac{\Delta P}{2\eta L}\cdot 2 = -\frac{\Delta P}{\eta L} \Longrightarrow (\partial_y^2+\partial_z^2)v_c = \emptyset$ Laplace equation

$v_c = Y(y)Z(z) \to (\partial_y^2+\partial_z^2)v_c = Y''Z + YZ'' = \emptyset \to \frac{Y''}{Y} = -\frac{Z''}{Z} = \lambda^2$

$Y'' - \lambda^2 Y = \emptyset \to (D^2-\lambda^2)Y = \emptyset \to D = \pm\lambda \to Y = c_1 e^{\lambda x} + c_2 e^{-\lambda x} = A\sinh\lambda x + B\cosh\lambda x$

$Z'' + \lambda^2 Z = \emptyset \to (D^2+\lambda^2)Z = \emptyset \to D = \pm i\lambda \to Z = C\sin\lambda z + D\cos\lambda z$

$v_c = (A\sinh\lambda y + B\cosh\lambda y)(C\sin\lambda z + D\cos\lambda z)$ $\quad v_c(y=\pm\frac{w}{2}, z) = \emptyset$

$z=0 \to v_c = (\dots)(0+D) = \emptyset \to \boxed{D=\emptyset}$ $\quad v_c(y, z=0) = v_c(y, z=h) = \emptyset$

$z=h \to v_c = (\dots)(C\sin\lambda h) = \emptyset \to C \neq \emptyset$ $\quad v_c(y,z) = v_c(-y,z)$

$\sin\lambda h = \emptyset \to \lambda h = n\pi \to \boxed{\lambda_n = \frac{n\pi}{h}}$

$v(y,z) = v(-y,z) \Rightarrow (A_n\sinh\lambda_n y + \cancel{B_n\cosh\lambda_n y})\cancel{(\dots)} = (A\sinh -\lambda_n y + \cancel{B_n\cosh -\lambda_n y})\cancel{(\dots)}$

$2A_n\sinh\lambda_n y = \emptyset \to \boxed{A_n=\emptyset} \Rightarrow \boxed{v_c = \sum_n B_n C_n \cosh\lambda_n y \sin\lambda_n z}$ $\quad B_n C_n = C'_n$

$v_x(y,z) = \frac{\Delta P}{2\eta L}(hz-z^2) + \sum_n C'_n \sin\lambda_n z \cosh\lambda_n y$

$y = \pm\frac{w}{2} \to \frac{\Delta P}{2\eta L}(hz-z^2) + \sum_n C'_n \sin\lambda_n z \cosh\lambda_n \frac{w}{2} = \emptyset$

$\sum_n C'_n \sin\lambda_n z \cosh\lambda_n \frac{w}{2} = +\frac{\Delta P}{2\eta L}(hz-z^2)$ expanding right side with Fourier Sine series

$+\frac{\Delta P}{2\eta L}(hz-z^2) = \sum_n b_n \sin\frac{n\pi z}{h}$, $\quad b_n = \frac{2}{h}\int_0^h \frac{+\Delta P}{2\eta L}(hz-z^2)\sin\frac{n\pi z}{h}\,dz$

$\Rightarrow \cancel{b_n = \frac{2}{h}\int_0^h \frac{+\Delta P}{2\eta L}}$ $\quad b_n = \frac{2}{h}\cdot\frac{+\Delta P}{2\eta L}\left[h\int_0^h z\sin\frac{n\pi z}{h}\,dz - \int_0^h z^2\sin\frac{n\pi z}{h}\,dz\right]$

$$\int_0^h z \sin\frac{n\pi z}{h}dz = \quad z=u \quad du=dz$$
$$dv=\sin\frac{n\pi z}{h}dz \rightarrow v=\frac{-h}{n\pi}\cos\frac{n\pi z}{h}$$

$$=-\frac{zh}{n\pi}\cos\frac{n\pi z}{h}+\int_0^h\frac{h}{n\pi}\cos\frac{n\pi z}{h}dz=-\frac{zh}{n\pi}\cos\frac{n\pi z}{h}+\frac{h^2}{n^2\pi^2}\sin\frac{n\pi z}{h}\Big)_0^h=\frac{-h^2}{n\pi}\cdot(-1)^n$$

$$\int_0^h z^2\sin\frac{n\pi z}{h}dz = \quad z^2=u \rightarrow du=2zdz \qquad \sin\frac{n\pi z}{h}dz=dv \rightarrow v=\frac{-h}{n\pi}\cos\frac{n\pi z}{h}$$

$$=-\frac{z^2h}{n\pi}\cos\frac{n\pi z}{h}+\int_0^h\frac{2h}{n\pi}z\cos\frac{n\pi z}{h}=-\frac{z^2h}{n\pi}\cos\frac{n\pi z}{h}+\frac{2h}{n\pi}\int z\cos\frac{n\pi z}{h}$$

$$\int_0^h z\cos\frac{n\pi z}{h}dz= \qquad u=z \quad du=dz \qquad dv=\cos\frac{n\pi z}{h}dz \quad v=\frac{+h}{n\pi}\sin\frac{n\pi z}{h}$$

$$=+\frac{zh}{n\pi}\sin\frac{n\pi z}{h}-\int\frac{h}{n\pi}\sin\frac{n\pi z}{h}dz=+\frac{zh}{n\pi}\sin\frac{n\pi z}{h}+\left(\frac{h}{n\pi}\right)^2\cos\frac{n\pi z}{h}$$

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$$h\int_0^h z\sin\frac{n\pi z}{h}dz-\int_0^h z^2\sin\frac{n\pi z}{h}dz=$$

$$\left[\frac{-h^3}{n\pi}\cdot(-1)^n-\left(-\frac{z^2h}{n\pi}\cos\frac{n\pi z}{h}+\frac{2h}{n\pi}\left(+\frac{zh}{n\pi}\sin\frac{n\pi z}{h}+\frac{h^2}{n^2\pi^2}\cos\frac{n\pi z}{h}\right)\right)\right]_0^h$$

$$=\left[\cancel{\frac{-h^3}{n\pi}\cdot(-1)^n}-\left(\cancel{-\frac{h^3}{n\pi}\cdot(-1)^n}+\frac{2h}{n\pi}\left(\cancel{\frac{-h^2}{n\pi}}\times 0+\frac{h^2}{n^2\pi^2}(-1)^n\right)\right)\right]-\frac{2h}{n\pi}\cdot\frac{h^2}{n^2\pi^2}$$

$$=\frac{-2h^3}{n^3\pi^3}\left((-1)^n-1\right) \qquad n\rightarrow\text{even}\Rightarrow\emptyset$$

$$n\rightarrow\text{odd}\Rightarrow +\frac{4h^3}{n^3\pi^3}\Rightarrow b_n=\frac{+2}{h}\cdot\frac{\Delta P}{2\eta L}\cdot\frac{4h^3}{n^3\pi^3} \qquad n\equiv\text{odd}$$

$$b_n=+\frac{4\Delta P}{\eta L}\cdot\frac{h^2}{n^3\pi^3}\rightarrow b_n=\frac{4\Delta P h^2}{\eta L\pi^3(2n+1)^3}$$

$$\frac{\Delta P}{2\eta L}(hz-z^2)=\sum_n\frac{4\Delta P h^2}{\eta L\pi^3(2n+1)^3}\cdot\sin\frac{(2n+1)\pi z}{h}$$

$$\sum_n c'_n\sin\lambda_n z\cosh\lambda_n w/2=\sum_m\frac{-4\Delta P h^2}{\eta L\pi^3(2n+1)^3}\cdot\sin\frac{(2n+1)\pi z}{h} \rightarrow$$ right side only having odd n terms $\Rightarrow$ left side must too.

$$\sum_n c'_n\cancel{\sin\frac{(2n+1)\pi z}{h}}\cosh\frac{(2n+1)\pi w}{2h}=\sum_n-\frac{4\Delta P h^2}{\eta L\pi^3(2n+1)^3}\cdot\cancel{\sin\frac{(2n+1)\pi z}{h}}$$

$$c'_n=\frac{-4\Delta P\cdot h^2}{\eta L\pi^3(2n+1)^3}\cdot\frac{1}{\cosh\left((2n+1)\pi\frac{w}{2h}\right)}$$

$$v_x = \frac{\Delta P}{2\eta L}(hz - z^2) - \sum_n \frac{4\Delta P .\, h^2}{\eta L \pi^3 (2n+1)^3} \cdot \frac{\cosh (2n+1)\pi y/h}{\cosh (2n+1)\frac{\pi w}{2h}} . \sin \frac{(2n+1)\pi z}{h}$$

$$Q = \int_{-w/2}^{w/2} dy \int_0^h dz\, v_x$$

$$\int_0^h dz (hz - z^2) = h . \frac{h^2}{2} - \frac{h^3}{3} = \frac{3h^3 - 2h^3}{6} = \frac{h^3}{6}$$

$$\int_{-w/2}^{w/2} \cosh \frac{(2n+1)\pi y}{h}\, dy$$

$$\int_0^h dz \sin\left(\frac{(2n+1)\pi z}{h}\right) = \frac{-h}{(2n+1)\pi} \cdot \cos\left(\frac{(2n+1)\pi z}{h}\right)\Big|_0^h$$

$$= \frac{-h}{(2n+1)\pi} \cdot (-2) = \frac{+2h}{(2n+1)\pi}$$

$$= 2\int_0^{w/2} dy \cosh \frac{(2n+1)\pi y}{h} = 2 . \frac{h}{(2n+1)\pi} \cdot \sinh\left(\frac{(2n+1)\pi y}{h}\right)\Big|_0^{w/2} = \frac{2h}{(2n+1)\pi} \sinh \frac{(2n+1)\pi w}{2h}$$

$$Q = \frac{\Delta P}{2\eta L} \cdot \left(\frac{h^3}{6}\right) \cdot (w) - \sum_n \frac{4\Delta P h^2}{\eta L \pi^3 (2n+1)^3} \cdot \frac{2h}{(2n+1)\pi} \cdot \tanh\left(\frac{(2n+1)\pi w}{2h}\right) \cdot \frac{2h}{(2n+1)\pi}$$

$$= \frac{\Delta P}{2\eta L} \cdot \frac{h^3 w}{6} - \sum_n \frac{16 \Delta P h^4}{\eta L \pi^5 (2n+1)^5} \cdot \tanh\left(\frac{(2n+1)\pi w}{2h}\right)$$

$$= \underbrace{\frac{\Delta P}{12\eta L} \cdot \frac{h^3 w}{1}} \left[1 - \sum_n \frac{192\, h}{w \pi^5 (2n+1)^5} \cdot \tanh\left(\frac{(2n+1)\pi w}{2h}\right)\right]$$

Parallel plate solution. $\frac{w}{h} \to \infty \Rightarrow \frac{h}{w} \to 0 \Rightarrow Q \to \frac{\Delta P h^3 w}{12\eta L}$